CS342 · VISUAL PROGRAMMING

Midterm 471 Practice

Find Output and writing a GUI

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01 Questions

Q1 FIND OUTPUT

Find the output of the program below: (a) with the input tab2, (b) with the input tab3.

```
public class average {
    private static int average(String[] list) throws ExceptionEmpty {
        int sum = 0, integer, nbNotes = 0;
        for (int i = 0; i < list.length; i++) {
            try {
                integer = Integer.parseInt(list[i]);
                if (integer < 0) throw new ExceptionRange("few");
                else if (integer > 20) throw new ExceptionRange("large");
                else {
                    sum += integer;
                    nbNotes++;
                }
            }
            catch (NumberFormatException nfe) {
                System.out.println("Note no." + (i+1) + " is not integr");
            }
            catch (ExceptionRange ei) {
                System.out.println("Note no." + (i+1) + " is too " + ei.getMessage());
            }
        } // Fin for
        if (nbNotes == 0) throw new ExceptionEmpty();
        return sum / nbNotes;
    }
    public static void main(String[] args) {
        String[] tab2 = {"-1", "2", "x", "30", "18"};     // input (a)
        String[] tab3 = {"x", " ", "2.3", "/n"};          // input (b)
        try {
            System.out.println("The average is: " + average(tab3));
        }
        catch (ExceptionEmpty er) {
            System.out.println(er);
        }
    }
}
```

```
public class ExceptionRange extends Exception {
    public ExceptionRange(String s) { super(s); }
}
public class ExceptionEmpty extends Exception {
    public String toString() { return ("No note is valid"); }
}
```

Q1 OUTPUT

(a) tab2:

(b) tab3:

Q2 FIND OUTPUT

Find the output of the program below.

```
public class Devices {
    void start() { System.out.println("Device is starting"); }
    void close() { System.out.println("Device is closing"); }
    class Phone extends Devices {
        @Override
        void start() { System.out.println("Phone is starting"); }
        void call()  { System.out.println("Phone is calling"); }
    }
    class Laptop extends Devices {
        @Override
        void start()   { System.out.println("Laptop is starting"); }
        void compile() { System.out.println("Laptop is compiling code"); }
    }
    public static void main(String[] args) {
        Devices d1 = new Phone();
        Devices d2 = new Laptop();
        ((Phone) d1).call();
        d2.start();
        ((Phone) d1).start();
        ((Laptop) d2).close();
        Devices d3 = new Devices();
        ((Phone) d3).call();
    }
}
```

Q2 OUTPUT

Output:

Q3 GUI

Write the Java class Q3 that displays the following login window (title "login"). Use a GridLayout of 3 rows, each row a JPanel with a FlowLayout.

